

AyB

P1] Sea $X \sim B_{100}(p)$ $H_0: p=0.4$ vs $H_1: p=0.5$

a) EMV per $\hat{p}_{EMV} = \frac{X}{100}$

$$C = \{x \in X: \hat{p}_{EMV} > 0.4\} \rightarrow \underset{\text{Decision}}{2} = \sup_{p \in \Theta_0} P(X \in C) \quad \text{Error Tipo I}$$

rejection de H_0 que H_0 es verdad

$$\begin{aligned} P_r(\{x \in X: X > 40\} | p=0.4) &= 1 - P_r(X \leq 40) \\ &= 1 - \sum_{k=0}^{40} \binom{100}{k} p^k (1-p)^{100-k}, \quad p=0.4 \\ &= 0.457 \end{aligned}$$

Error Tipo II

$$\begin{aligned} \beta &= P_r(X \notin C | H_0 \text{ es falsa}) \\ &= 1 - P_r(X \in C | H_0 \text{ falsa}) \\ &= P_r(X \leq 40 | p=0.5) \\ &= 0.028 \end{aligned}$$

b) Test MP Lema Neyman - Pearson

$$\Lambda(\theta_0, \theta_1) = \frac{L(\theta_0)}{L(\theta_1)} = \frac{\binom{100}{x} p_0^x (1-p_0)^{100-x}}{\binom{100}{x} p_1^x (1-p_1)^{100-x}}$$

$$= \left(\frac{p_0}{p_1}\right)^x \left(\frac{1-p_0}{1-p_1}\right)^{100-x} = \left(\frac{4/10}{5/10}\right)^x \left(\frac{6/10}{5/10}\right)^{100-x}$$

$p_0 = 0.4$
 $p_1 = 0.5$

$$C^* = \{x \in X: \Lambda(\theta_0, \theta_1) \leq K_\alpha\}$$

$$C^* = \left\{x \in X: \left(\frac{2}{3}\right)^x \leq K_\alpha\right\}$$

$x \ln\left(\frac{2}{3}\right) \leq \ln K_\alpha \rightarrow$

$$C^* = \{x \in X: x \geq k_\alpha^{**}\}$$

$$\begin{aligned} &= \left(\frac{4}{5}\right)^x \left(\frac{6}{5}\right)^{100-x} \\ &= \left(\frac{4}{5}\right)^x \cdot \left(\frac{6}{5}\right)^{100} \\ &= \left(\frac{2}{3}\right)^x \cdot C \end{aligned}$$

(1)

$$C^* = \{x \in X: X \geq k^*\}$$

$$Pr(x \in C^* | H_0) = \alpha$$

$$Pr(\{x \in X: X \geq k^*\} | H_0) = \alpha, \quad X \sim B_{100}(100, p_0)$$

$$Pr(X \geq k^*) = \alpha$$

$$1 - Pr(X \leq k^* - 1) = \alpha$$

$$1 - \alpha = Pr(X \leq k^* - 1)$$

$$0.95 = Pr(X \leq \underline{k^* - 1})$$

$$48 \rightarrow k^* = 49$$

$$C^* = \{x \in X: X \geq 49\}$$

MP

Error Tipo II

$$\beta = Pr(X \notin C^* | H_1)$$

$$= Pr(X < 49 | H_1)$$

$$= Pr(X \leq 48 | H_1)$$

$$\beta \approx 0.3021$$

Error Tipo I

$$\alpha = Pr(X \in C^* | H_0)$$

~~no es~~

$$C^* = \{x \in X: X \geq 49\}$$

El primer TIS propuesto tiene un ~~mayor~~ tamaño (mayor error Tipo I) por lo que la ineficiencia del estimador.

P2) $X_1, \dots, X_n$ i.i.d. $p(x|\theta) = \binom{3+x}{x} \theta^4 (1-\theta)^x$ $X=0,1,2, \dots$
 $X \sim NB(4, \theta)$ $0 < \theta < 1$

(a) Test MP de tamanho 2 $H_0: \theta = 0.8$
 $H_1: \theta = 0.2$

$$\Lambda(\theta_0, \theta_1) = \frac{L(\theta_0)}{L(\theta_1)}, \quad L(\theta) = \frac{\prod_{i=1}^n \binom{3+x_i}{x_i} \theta^{4n} (1-\theta)^{\sum x_i}}{g(x_i)}$$

$$= \frac{g(x_i) \theta_0^{4n} (1-\theta_0)^{\sum x_i}}{g(x_i) \theta_1^{4n} (1-\theta_1)^{\sum x_i}}$$

$$= \left(\frac{\theta_0}{\theta_1} \right)^{4n} \left(\frac{1-\theta_0}{1-\theta_1} \right)^{\sum x_i}$$

$$= \left(\frac{8/10}{2/10} \right)^{4n} \left(\frac{2/10}{8/10} \right)^{\sum x_i}$$

$$= (4)^{4n} \cdot \left(\frac{1}{4} \right)^{\sum x_i}$$

$B_{(n+m), N_y, j_n}$
 $p(x|\theta) = \binom{r+x-1}{x} \theta^r (1-\theta)^x$
 $r=4$
 $\theta=\sigma$

$$C^* = \left\{ x \in \mathcal{X} : 4^{-\sum x_i} \leq K_2^* \right\}$$

$$- \sum x_i \leq K_2^{**}$$

$$C^* = \left\{ x \in \mathcal{X} : \sum x_i \geq K_2^{***} \right\}$$

$$P_r(\sum x_i \geq K_2^{***} | H_0) = 2, \quad K_2^{***} \text{ adequado}$$

$\sum x_i \sim NB(4 \cdot 20, 0.8)$
 $\sum x_i \sim NB(80, 0.8)$

$$1 - P_r(\sum x_i \leq K_2^{**} | H_0) = 2$$

b) $n=20, \alpha=0.01$

$$0.99 = P_r(\sum_{i=1}^{20} x_i \leq \underline{K_2^{**}-1} | H_0)$$

$\Rightarrow K_2^{**} = 34$

$$C^* = \left\{ x \in \mathcal{X} : \sum x_i \geq 34 \right\}$$

P3 $X \sim \text{Geom}(p)$

$H_0: p \geq 0.5$
 $H_1: p < 0.5$

a) Test UMP 0.05
 $\theta_1 < \theta_2$

$$\Lambda(\theta_1, \theta_2) = \frac{\theta_1 (1-\theta_1)^{x-1}}{\theta_2 (1-\theta_2)^{x-1}}$$

$$= \frac{\theta_1}{\theta_2} \left(\frac{1-\theta_1}{1-\theta_2} \right)^{x-1}$$

$1-\theta_1 > 1-\theta_2$ $\downarrow$
 es creciente
 e $\frac{1-\theta_1}{1-\theta_2} > 1$ respecto a X

Karlin-Rubi (Apéndice 4.1)

$C^* = \{x \in X: X \geq K_2\} = 2$

$\sup_{\theta \in \Theta_0} P_r(X \in C^*) = 2$

$\sup_{\theta \in \Theta_0 = [0.5, 1]} 1 - P_r(X \leq K_2 - 1) = 2 = 0.05$
 monoton decreasing

$1 - P_r(X \leq K_2 - 1 | \theta = 0.5)$

$0.95 = P_r(X \leq \underline{K_2 - 1} | \theta = 0.5)$

$4 \rightarrow K_2 = 5$

$C^* = \{x \in X: X \geq 5\}$

Teorema Karlin-Rubi

$NM \Rightarrow \Lambda(\theta_1, \theta_2) = \frac{L(\theta_1)}{L(\theta_2)}$ es $\uparrow$ estrictamente de $T(\bar{X})$

i) si NM no decreciente
 $\rightarrow$ elegir crítica UMP de Tammara
 a por probar

$H_0: \theta \leq \theta_0$

$H_1: \theta > \theta_1$

es $C^* = \{x \in X: T(x) \geq K_\alpha\}$

ii) NM no creciente

$H_0: \theta \leq \theta_0$

$H_1: \theta > \theta_1$

$C^* = \{x \in X: T(x) \leq K_\alpha\}$

$2 = P_r(X \geq K_2) = P_r(X > K_2 - 1) = (1-p)^{K_2}$
 decreciente a p

$P_r(X > c) = (1-p)^{c+1}$

b) $X=5 \rightarrow$ Rechazar $H_0 \rightarrow$ La moneda es la cargada

PM) $X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta) = \frac{3x^2}{\sigma^3} \exp\left\{-\frac{x^3}{\sigma^3}\right\} \mathbb{1}_{(0,\infty)}(x)$

a) Test UMP τ_{α} and p -value. $H_0: \sigma \geq 2$
 $H_1: \sigma < 2$

$$\Lambda(\sigma_1, \sigma_2) = \frac{L(\sigma_1)}{L(\sigma_2)} = \frac{\prod_{i=1}^n \frac{3x_i^2}{\sigma_1^3} \exp\left\{-\frac{x_i^3}{\sigma_1^3}\right\}}{\prod_{i=1}^n \frac{3x_i^2}{\sigma_2^3} \exp\left\{-\frac{x_i^3}{\sigma_2^3}\right\}}$$

$$= \left(\frac{\sigma_2}{\sigma_1}\right)^{3n} \exp\left\{-\sum_{i=1}^n x_i^3 \left(\frac{1}{\sigma_1^3} - \frac{1}{\sigma_2^3}\right)\right\}$$

≥ 0

No monotone can respect $\sum x_i^3$

Kolmogorov Smirnov (T-test)

$$C^* = \{x \in \mathcal{X} : \sum x_i^3 \leq K_\alpha\}$$

$$= \{x \in \mathcal{X} : \frac{\sigma^3}{2} \sum z_i \leq K_\alpha\}$$

$$= \{x \in \mathcal{X} : \chi_{2n}^2 \leq K_\alpha \cdot \frac{2}{\sigma^3} = 2\}$$

$$\alpha = \sup_{\sigma \in \Theta_0 = (2, \infty)} P_\sigma(X \in C)$$

$$\alpha = P_\sigma(X \in C | \sigma=2)$$

$$= P_\sigma(\chi_{2n}^2 \leq K_\alpha \cdot \frac{1}{4}) = 2$$

$$K_\alpha \cdot \frac{1}{4} = \chi_{2n}^2(\alpha)$$

$$C^* = \{x \in \mathcal{X} : \sum x_i^3 \leq 4 \chi_{2n}^2(\alpha) | K_\alpha = \chi_{2n}^2(\alpha) \cdot 4\}$$

$$Y_i = X^3 \quad \left| \frac{d}{dx} \right.$$

$$\frac{dy}{dx} = 3x^2 dx$$

$$Y_i \sim \exp(1/\sigma^3)$$

$$f(y|\sigma) = \frac{1}{\sigma^3} \exp\left\{-\frac{y}{\sigma^3}\right\}$$

$$Z = \frac{Y}{\sigma^3} \sim \exp(1)$$

$$dx = \frac{dy}{\sigma^3}$$

$$Z_i = \frac{2X_i^3}{\sigma^3} \sim \exp(1/2)$$

$$\sum Z_i \sim \text{Gamma}(n, 1/2) \\ \sim \chi_{2n}^2 \quad \textcircled{2}$$

b) Supponi che esse Tipo de Test
 no existe un Test UMP semplice
 e per ello questo e' il
 Test de rank de verosimilitud

$$H_0: \sigma = 2$$

$$H_1: \sigma \neq 2$$

$$\Lambda(x) = \frac{\sup_{\sigma \in \Theta_0} L(\sigma)}{\sup_{\sigma \in \Theta} L(\sigma)}$$

$$H_0: \sigma \in \Theta_0$$

$$H_1: \sigma \in \Theta_1$$

$$C_\alpha = \{x \in X: \Lambda(x) \leq k_\alpha\}$$

$$\Lambda(x) = \frac{3^n (\prod x_i)^2 \exp\left\{-\frac{\sum x_i^3}{\sigma}\right\}}{\sigma^n}$$

$$\frac{3^n (\prod x_i)^2 \exp\left\{-\frac{\sum x_i^3}{\hat{\sigma}}\right\}}{\hat{\sigma}^{3n}}$$

$$= \frac{\hat{\sigma}^{3n}}{\hat{\sigma}^{3n}} \exp\left\{-\frac{\sum x_i^3}{\hat{\sigma}} + n\right\}$$

$$= \frac{(\sum x_i^3)^n}{(\hat{\sigma})^n} \exp\left\{-\frac{\sum x_i^3}{\hat{\sigma}} + n\right\}$$

$$s(\tau) = \left(\frac{\hat{\sigma}_{ENV}}{2}\right)^{3n} \exp\left\{n - n\left(\frac{\hat{\sigma}_{ENV}}{2}\right)^3\right\}$$

$$\tau = \frac{\sum x_i^3}{2}$$

$$s(\tau) = \tau^{3n} \exp\{n - n\tau^3\}$$

$$s'(\tau) = 3n \tau^{3n-1} \exp(n - n\tau^3) - n \tau^{3n} \exp(n - n\tau^3) \cdot 3\tau^2$$

$$= 3n \tau^{3n-1} \exp(n - n\tau^3) (1 - \tau^3)$$

$$s'(\tau) > 0, \tau < 1$$

$$s'(\tau) < 0, \tau > 1$$

$\tau = 1$ e
 massimo

$$L(\sigma) = \frac{3^n (\prod x_i)^2}{\sigma^{3n}} \exp\left\{-\frac{\sum x_i^3}{\sigma}\right\}$$

$$l(\sigma) = 3 \ln(3) + 2 \sum \ln(x_i) - 3n \ln(\sigma) - \frac{\sum x_i^3}{\sigma}$$

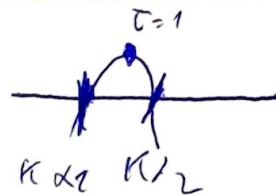
$$\frac{dl(\sigma)}{d\sigma} = -\frac{3n}{\sigma} + \frac{\sum x_i^3}{\sigma^2} = 0$$

$$\frac{\sum x_i^3}{\sigma^2} = \frac{n}{\sigma}$$

$$\left(\frac{\sum x_i^3}{n}\right)^{1/3} = \hat{\sigma}_{ENV}$$

Per $1/\sigma \sqrt{t}$

$$\frac{\sigma_{Emv}^2}{2}$$



$\Lambda(\sigma)$ $\bar{t}_{mv}$ $\sim$ máxima $\sim \bar{t}=1$ $\rho \sqrt{t}$ ρ

$$\Lambda(x) \leq K_2 \Leftrightarrow \cancel{\Lambda(\sigma=2)} \quad \frac{\sigma_{Emv}^2}{2} \leq K_{d1} \quad \text{or} \quad \frac{\sigma_{Emv}^2}{2} \geq K$$

Dado

$$Pr(\hat{\sigma}_{Emv} \leq 2 K_{d1} | \sigma=2) = d_1 \quad \wedge \quad Pr(\hat{\sigma}_{Emv} \geq 2 K_{d2} | \sigma=2) = d_2$$